

CASE STUDY PRACTICAL

1.A user reports their laptop won't power on even when plugged in. List 5 troubleshooting steps you would take to diagnose the cause, and for each step, explain what you are checking for.

ANS:

Troubleshooting Action	What You Are Checking
1.Check the wall socket	Confirm that the power outlet is working properly.
2.Inspect the charger and cable	Look for cuts, bent pins, loose connections, or physical damage.
3.Check the charging indicator	Look for cuts, bent pins, loose connections, or physical damage.
4.Check the charging port	Look for dust, damage, or a loose port that may prevent power from reaching the laptop.
5.Test the charger with a multimeter	Measure the adapter's output voltage and compare it with the voltage printed on the charger label.

2.You receive a helpdesk ticket: 'My monitor is showing a blank screen, but the CPU seems to be running.' Write a step-by-step troubleshooting flow you would follow to identify and resolve the issue.

Hint: Consider checking both hardware connections and basic display settings.

ANS :

Blank Monitor Screen – Troubleshooting Flow

1. **Check the monitor power** – Make sure the monitor is switched on and the power LED is lit.
2. **Check the display cable** – Ensure the HDMI/DisplayPort/VGA cable is firmly connected at both ends.
3. **Check the correct input source** – Select the input (HDMI/DP/VGA) that matches the connected cable.
4. **Check the GPU connection** – Make sure the graphics card is properly seated and its power cable is connected.
5. **Test another cable/monitor** – Use a known-working cable or monitor to identify whether the problem is with the display equipment.
6. **Check RAM** – Power off the PC and reseal the RAM if there is still no display.
7. **Check display settings** – If Windows starts but the screen remains blank, try Win + Ctrl + Shift + B to reset the graphics driver.
8. **Restart and confirm** – Restart the PC and verify that the display works normally.

Likely causes: Loose display cable, incorrect monitor input, faulty GPU, faulty RAM, or a defective monitor/cable.

3. A user complains that their wireless mouse is unresponsive, but the keyboard works fine. Describe how you would isolate whether the issue is with the mouse, the USB port, or the system settings.

ANS:

Wireless Mouse Not Working – Troubleshooting Steps

1. **Check the mouse power** – Make sure the mouse is switched on and the battery is charged/working.
2. **Check the USB receiver** – Remove and reconnect the wireless receiver firmly.
3. **Try another USB port** – Connect the receiver to a different USB port to check whether the original port is faulty.
4. **Test the mouse on another PC** – If it works there, the mouse is probably fine and the issue is with the original PC.
5. **Test another mouse on the PC** – If another mouse works, the original mouse is likely faulty.
6. **Check system settings/drivers** – Open **Device Manager** → **Mice and other pointing devices** and check for driver errors; update or reinstall the mouse driver if needed.
7. **Confirm the result** – If the receiver works in another USB port and the mouse works on another PC, check the original system's USB/driver settings for the cause.

4. Imagine you are working on a helpdesk team for a gaming cafe. One system is overheating and shutting down during use. List 3 possible hardware causes and describe how you would check each one.
Hint: Think about fans, dust, and thermal paste.

ANS:

Possible Hardware Cause	How to Check
1. Faulty or stopped cooling fan	Open the case and check whether the CPU/GPU/case fans are spinning properly when the PC is running.
2. Dust buildup	Inspect the fans, heatsink, and air vents for dust; clean them carefully using compressed air.
3. Dried or poorly applied thermal paste	Check the thermal paste between the CPU and heatsink; if it is dried out, clean and replace it with fresh thermal paste.

